



A Systematic Analysis of Construction Performance Management: Key Performance Indicators from 2000 to 2020

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Abstract

Today, the construction industry has to turn into one of the major industries, which covers a wide range of dependent industries. Rapid changes in the construction environment require a great deal of effort for a company or organization to maintain survival and growth. The life of companies and organizations in the field of construction depends on the way they manage their projects. Achieving project success is a major goal of project managers. A method of evaluating and controlling the project success is to apply project performance management. As a result of the studies, the most important key performance indicators, their relevant sources and references, and the relationships related to the quantification of indicators and their relationships with each other have been introduced and described. These results could help researchers in the future to obtain information and resources related to each index. Safety, environment, cost, profitability, scheduling, productivity, sustainability, quality, client satisfaction, and team satisfaction are the most important indicators.

Keywords Performance management · Construction projects · Key performance indicators · Systematic literature review

1 Introduction

The construction industry is one of the mother industries of each country, which is comprised of a wide range of related industries. This industry plays a crucial role in the economic growth of every country. Therefore, the complexity of construction projects is always increasing. This complicity could be due to a wide variety of reasons, including the rapid development in tools and engineering skills, the rising costs of raw materials, the rise in stakeholder expectations, and the like. Today, the number of people involved in construction teams is increasing, with contractors obliged to have a large number of experts on their side. Under such conditions, managing these people and complexities will be difficult.

The main goal of project managers is to ensure the success of construction projects. However, the definition and evaluation of success in this regard have been the subject of discussion for many years. Since the construction industry is a key element of the state's development and progress, particular attention must be paid to achieving success in implementing projects in this industry (Mohammadi et al. 2018). Although the definition of project success differs from the perspective of every stakeholder, it is in general defined based on achieving project goals and expectations. One way to evaluate project success is to manage project performance. Project performance is a subset of project success (Fung and Siow 2013). The success of a construction project depends on the way the project team could work so effectively as to achieve project objectives within the specified scope, cost, and quality (Parikh and Phugat 2019).

The accurate and reliable prediction of project performance is critical for the success of construction projects. Such a prediction could be considerably effective in issuing an early warning of potential problems. In the construction industry, project managers must be able to make reliable predictions status of projects. Such predictions must help contractors control projects during the construction phase and give them an early warning of potential problems (Leon et al. 2017).

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Project success evaluation is also a useful tool in the construction industry employed to manage, control, and improve policies and predict the future success of projects (Chovichien and Nguyen 2013).

Determining various aspects of the concept of project success is difficult because projects are complex and dynamic. The definition of this concept could vary depending on the type of industry, the project team, or the individual perspective (Parfitt and Sanvido 1993). An architect may evaluate success in terms of aesthetics, an engineer in terms of technical competence, an accountant in terms of budgeted costs, a human resources manager in terms of employee satisfaction, and an executives officer in terms of market shares (Freeman and Beale 1992).

In the past years, project success was simply defined solely based on the implementation stages of the project life cycle. However, it is now required that project success be defined from the beginning to the end of the project life cycle (Iram et al. 2016).

Performance management is the continuous process of identifying, measuring, and developing performance in organizations by linking every individual's performance and goals to the overall organizational goals and mission. The underlying assumption is that by managing the performance of an individual and that of a team, departmental and organizational performance will be optimized; also, organizational performance improves upon an improvement in the levels of individual and team performance. In contrast, when the performance of an individual or a team is not managed, it will lead to frustration and dissatisfaction among team members. Performance management should lead to positive outcomes, and if done correctly, it will be a very effective tool for improving employee productivity and progress (Parikh and Phugat 2019).

Therefore, in this research, we try to identify and introduce major performance measurement systems and frameworks as well as the key performance measurement indicators, by reviewing the research literature on project performance measurement. In the case of the success of our attempts, a step will be taken in improving the performance of construction projects.

2 Literature Review

The main research questions include the following:

1. What are the most important key indicators used in research in recent years?
2. What is the priority of the importance of key indicators?
3. What are the interactions and the effects of the indicators on each other?

The following databases were then identified for search purposes:

- Scopus
- Science Direct
- John Wiley & Sons
- Springer
- etc.

We also searched the titles, abstracts, and keywords of the articles by searching below:

Project Management
Performance Management and construction project
Key Performance Indicator
Safety and Project Management
Cost and Project Management
Schedule and Project Management

etc.

3 Related Work

Project management is defined based on the overall success of the project's goals and expectations. These goals and expectations include technical, financial, educational, social, and professional aspects. Project success is the basis for managing and controlling current projects and planning for future ones (Chovichien and Nguyen 2013).

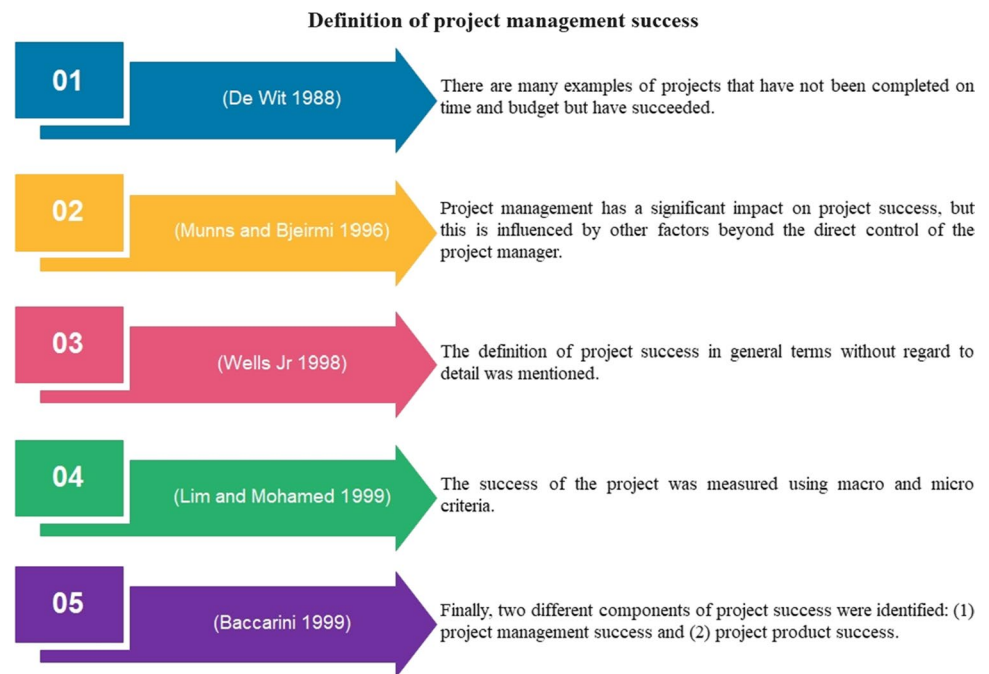
It will be stated in this research that project success is divided into the two components of (1) project management success and (2) project product success, which will be implied in Fig. 1.

A comprehensive model was developed to evaluate the success of construction projects, taking into account the two components of project success, namely project management and project product (Heravi and Ilbeigi 2012). In this study, critical indicators for each component have been considered as follows:

1. Project product success criteria: profitability, product quality, client satisfaction, contractor professional satisfaction, and investment performance.
2. Project management success criteria: cost, billing, scheduling, process quality, safety, and environment.

In the end, a function is defined for the success of the project product that takes into account the final product of the project as well as a function for the success of project management, and it also considers the final results of the way a project is managed. According to the results of this

Fig. 1 The evolution of the project success definition into the two components of product success and project management success



the quantitative model presented in this study, in addition to subjective and expert judgment, could be used as a reliable tool for contractors to ensure project success.

Various criteria have been used to evaluate project success for the past years. Traditionally, the three criteria of cost, time, and quality have been used for this purpose. However, many scholars believe that success could not be measured solely based on these three criteria because project success is a very complicated issue (Kennerley and Neely 2002).

By reviewing the results of past research, it could be concluded that the Iron Triangle of project management is no longer responsive to all project managers needs for comprehensive project performance control. The reason is that in today's complex projects, other performance aspects, such as safety, environment, and stakeholders, need to be evaluated by the project manager (Ogunlana 2010).

A group of researchers argued that the three criteria of time, cost, and quality are insufficient. Hence, other factors must be considered, such as the quality of the relationships among the parties and the flexibility in incorporating changes that could affect client satisfaction, thereby leading to project success or failure (Ward et al. 1991).

Since 1980, more emphasis has been put on non-financial and multidimensional performance indicators to better understand and manage the performance of construction projects (Excellence 2006; Leong et al. 2014; Neely et al. 2000; Rankin et al. 2008).

Also, some researchers have classified key performance indicators into quantitative and qualitative indicators (Chan and Chan 2004). Other researchers evaluated the impact

of rework on project performance from the perspective of employers and contractors (Hwang et al. 2009). Later on, the effects of delays on the delivery of construction projects and the project performance were examined (Owolabi James et al. 2014). These effects include increased costs, increased time, reduced productivity of human resources and profits, as well as a project failure. In 2014, the integrated performance evaluation of construction projects was practically examined. In their research, performance indicators including the status statement, scheduling, profitability, costs, quality, safety, customer satisfaction, and team satisfaction were taken into account, and finally, a general index was introduced for project performance evaluation (Nassar and AbouRizk 2014). In other research, the performance criteria of construction projects were evaluated according to project implementation phases (Yun et al. 2016). Two researchers evaluated and modeled the competence of construction projects and their relationships with project performance (Omar and Fayek 2016). At another time, the system dynamics approach was introduced to model the relationship between the bidding strategy and the performance of construction projects (Wibowo et al. 2017). In the same year, a project model was developed to predict the performance of construction projects using a system dynamics approach to managing project performance (Leon et al. 2017).

In 2018, a study was conducted in South Korea on measuring project performance in terms of optimal best management practices in the field of building construction. This study presented an algorithm using which project stakeholders could effectively measure and analyze the performance level of a building project together with project

characteristics and identified BMPs. The system this study offered was a means of easily quantifying project performance data to propose the optimal management method based on the relevant data management process (Cha and Kim 2018). In the same year, a review of the factors affecting safety performance in construction projects was conducted. In this research, a hierarchical framework was developed to show how the factors extracted affect safety performance in construction projects (Mohammadi et al. 2018).

In 2019, a study was conducted on evaluating the performance of construction organizations using a fuzzy preference relation technique. This study was conducted to examine various factors and attributes required for performance measurement of construction organizations operating in developing countries, such as India (Tripathi et al. 2019).

Making predictions is an important aspect in the field of project management. Accordingly, numerous studies have predicted the performance of construction projects. Value management is one of the most prominent methods, being widely used as a standard project control tool in the construction industry. However, due to the behavior and weaknesses of the EVM performance indicators of scheduling, it has been criticized by many researchers (Henderson 2007; Lipke 2003). In addition, it has been proven that EVM indicators are unreliable, so they do not suit projects with a nonlinear cumulative cost curve (Corovic 2006). Besides, the planned value method yields unreliable and poor results for projects continuing after the scheduled completion date (Henderson 2007). Figure 2 shows other methods used to predict performance (Al-Tabtabai 1998; Barraza et al. 2004; Farghal and Everett 1997; Ingle and Mahesh 2020; Li 2004; Nassar 2005; Tripathi et al. 2019).

Various methods have been employed over the years for predicting the performance of construction projects, with each of which having advantages and disadvantages. Among these methods, the dynamic system method has been used frequently together with other methods. In the present

research, the dynamic system method has been investigated and introduced as one of the best performance forecasting methods in construction projects. System dynamics (SD), developed by Professor Forster in 1958, is a well-known method employed for recognizing, visualizing, and analyzing complex dynamic feedback systems (Sterman 2000). System dynamics examines behaviors of the system from a general perspective by specifically focusing on interactions between elements of the system (Ding et al. 2016). In addition, it is a set of conceptual tools making it possible to understand the structure and dynamics of complex systems; besides, it ultimately gives the possibility of formulating more effective policies by careful modeling and computer simulation (Sterman 2000). System dynamics has been reported as a modeling tool used in the field of project management (Sterman 1992).

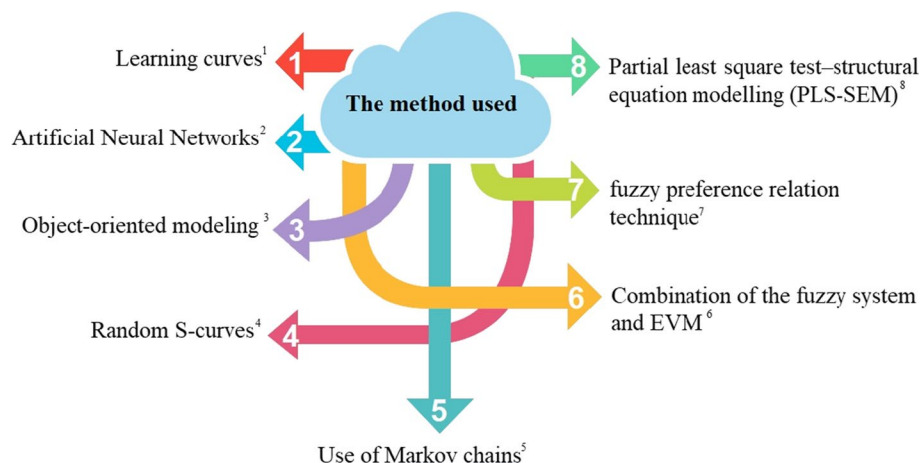
Figure 3 demonstrates a variety of applications utilized in recent years for system dynamics modeling in improving project performance management.

Also, one of the studies conducted in this research is related to journals in which functional management articles have been published. The articles reviewed in this study have been published in various prestigious journals. Journals that publish performance management topics are shown in Fig. 4, depending on the number of related articles available.

4 Methodology

To answer the questions posed at the beginning of the study, this study attempts to use a systematic and library review method to provide credible sources that discuss performance management, key performance indicators, forecasting systems and frameworks and size. Performance capture to be paid. While systematic reviews are often applied in the biomedical or healthcare context, they can be used in other areas where an assessment of a

Fig. 2 Models used to predict the performance of construction projects. Farghal and Everett (1997), Al-Tabtabai (1998), Li (2004), Barraza et al. (2004), Nassar (2005), Naeni et al. (2014), Tripathi et al. (2019), Ingle and Mahesh (2020)



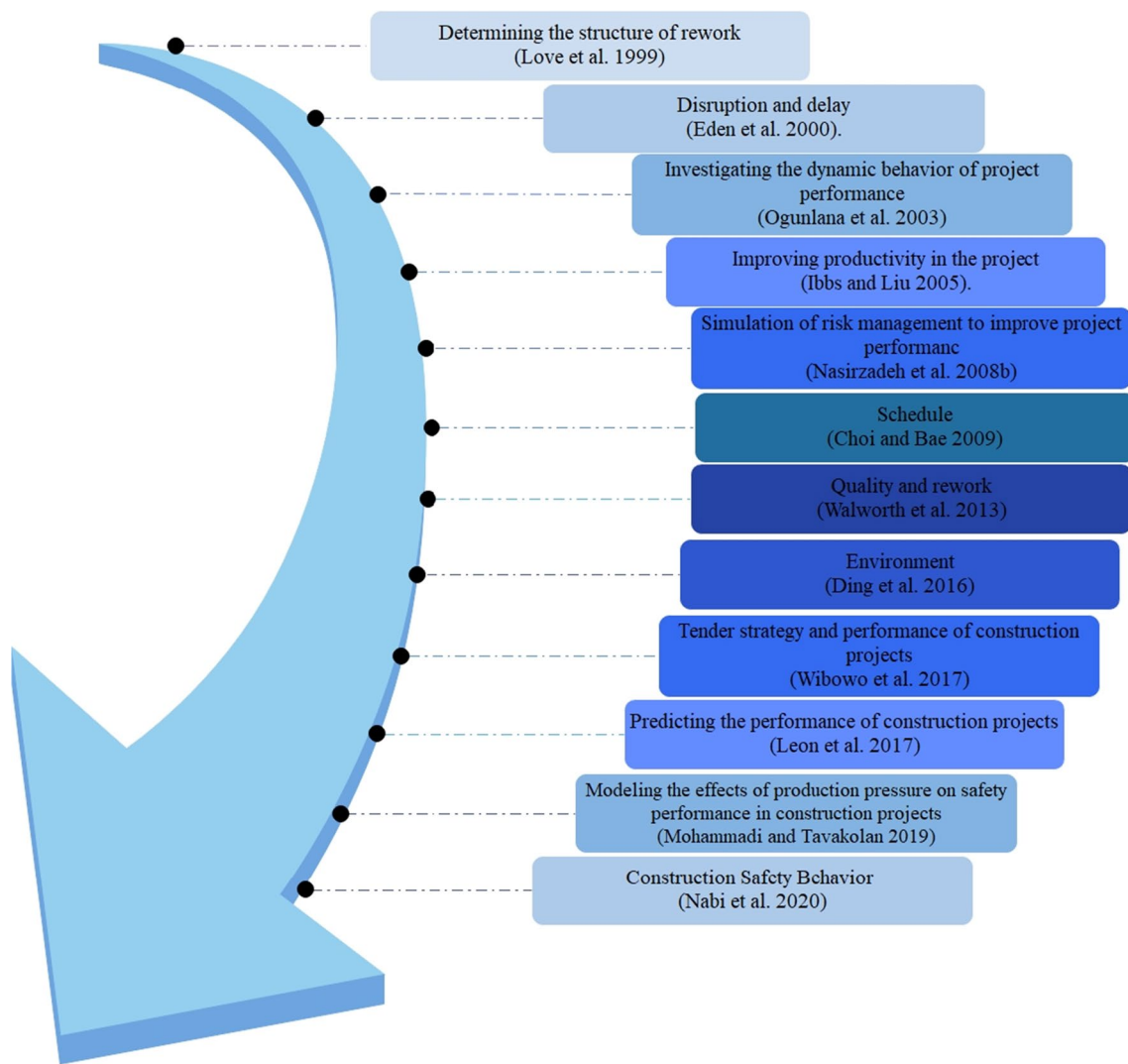
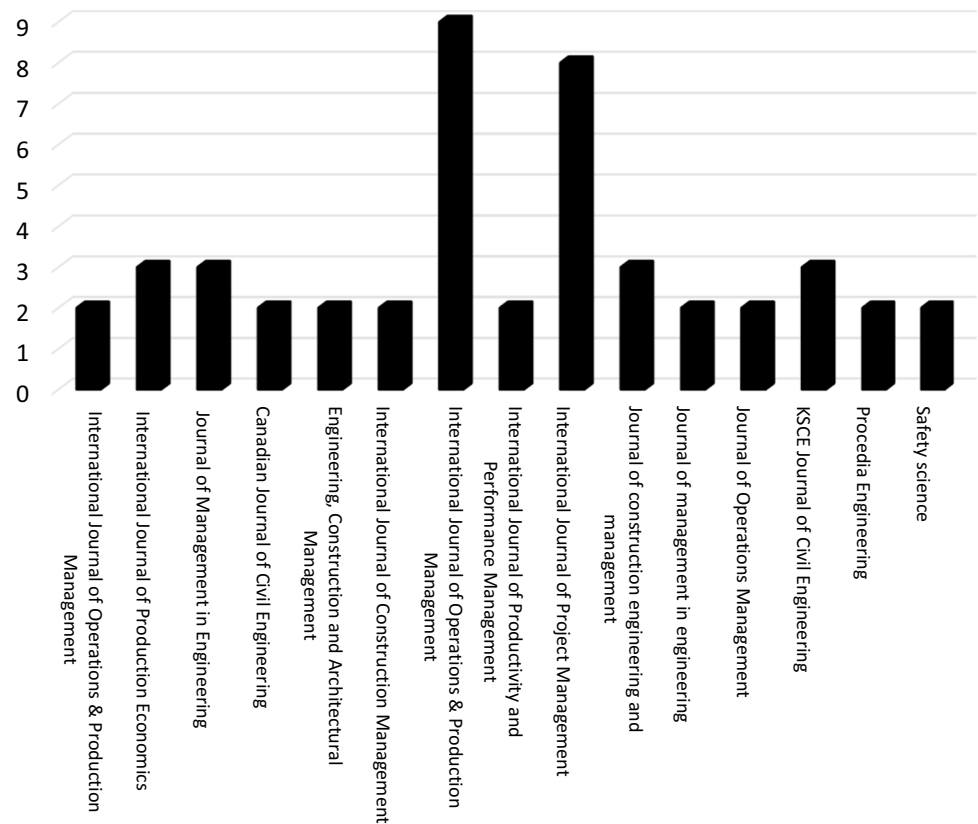
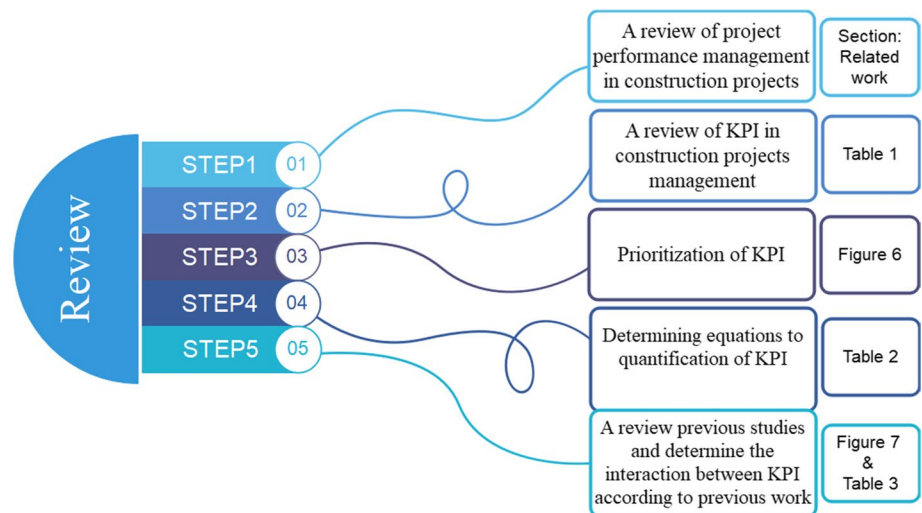


Fig. 3 System dynamics modeling in improving project performance management

precisely defined subject would be helpful (Adèr 2008). Systematic literature review is done by providing a clear, accurate, reproducible, and measurable method for evaluating and analyzing all available research on a particular research question, subject matter, or other phenomena (Oates and Capper 2009). Because individual studies can be scattered and represent only one aspect of a larger image, the value of a systematic review is that it is a combination of discrete parts and combines the results in an organized way, allowing you to have an overview of the research topic (Cooper 1986). Given that this research is a review of the literature on performance management and system dynamics in recent years, the research method is following the framework shown in Fig. 5.

5 Key Performance Indicators

Criteria measuring project success or failure are often referred to as "key performance indicators," or performance dimensions (Atkinson 1999; Beatham et al. 2004; Chan and Chan 2004; Shenhar and Wideman 1996). It is stated that during the early years of the project performance management review, one must know that "one cannot manage what they cannot measure" (Sanvido et al. 1992). In addition, researchers have stated, "if you cannot measure it, how can you improve it?" (Shen et al. 2000). Therefore, determining key performance indicators to evaluate the performance of construction projects to properly

Fig. 4 Journals that publish performance management topics**Fig. 5** A research framework

manage their performance in line with the set objectives is the main goal of project performance management. In construction projects, key performance indicators are considered appropriate and effective tools for determining the success rate of any projects or organizations. Therefore, by taking advantage of them, it would be possible to get one step closer to the desired goals. Besides, these indicators are basic information required for monitoring and controlling projects and ensuring that key objectives are achieved

(Leon et al. 2017). Research in 2009 showed that determining key performance indicators (KPIs) is the major step in establishing an appropriate performance measurement framework (Skibniewski and Ghosh 2009). However, there are already some performance indicators that can be used to simultaneously verify and enhance the performance of projects and organizations. Nevertheless, there are certain indicators among these indicators, which play a fundamental role in the overall performance of projects

and organizations; these indicators are called key performance indicators (KPIs), as already stated. However, these indicators, falling into different categories, may vary from project to project. Quantitative and qualitative indicators are of these types of classifications.

In all studies, a group of performance indicators have been reviewed and evaluated, and a summary of these studies is given in Table 1.

In this study, according to the previous explanations and the studies classified according to Table (4), indicators in recent years have been ranked as Fig. 6 shows.

In addition, the key performance indicators obtained at the previous stage were ranked by distributing a questionnaire, with the following results obtained for the most important indicators:

1) safety, 2) environmental, 3) cost, 4) profitability, 5) schedule, 6) productivity, 7) quality, 8) client satisfaction, 9) team satisfaction.

At this stage, definitions of each KPI with its application in research on performance evaluation and improvement were provided. In the end, the relationships between their quantifications and references were elucidated.

5.1 Cost Performance Indicator

The CPI (Cost Performance Index) is merely a measure of the completion of a construction project according to the projected budget. This index is determined by comparing the current costs allocated to the work and the budgeted costs allocated to the completed work up to the measurement date (Vyas and Kulkarni 2013). Cost performance has been proven to be a measure of project performance associated with client satisfaction (Idrus et al. 2011). Timely and quality delivery of a project according to the set budget is very important. Besides, the CPI is a measure of cost-effectiveness in projects.

5.2 Scheduling Performance Indicator

The SPI is a very important index in construction projects. A project will be successful in terms of scheduling performance if it is completed according to the time specified in the contract or before that time, according to the required quality (Dawood et al. 2006). When a project is delayed, it can make the client and stakeholders bear additional costs, thereby inflicting damage to them. Clients, consultants, and contractors consider scheduling performance as one of the main success criteria in projects. Accordingly, they have arrived at the conclusion that scheduling performance affects other factors, such as cost, quality, as well. The importance of the SPI in measuring the performance of construction projects in terms of client satisfaction has been proven. In addition, it has been stated to play a significant

role in assessing the quality management system of projects (Leong et al. 2014).

5.3 Quality Performance Indicator

Quality is the third constituent of the traditional performance indicator of the "Iron Triangle" or the "Golden Triangle," which is assessed according to the time and cost of the projects. Quality requirements of construction projects are stipulated in the contracts from the beginning. Other parts of these requirements and information could be found in other supplementary documents, such as orders. Therefore, a quick way to measure quality performance in a project is to compare the final product with the requirements in project documentation. This documentation specifies the competence and experience of employees, quality of materials and equipment, quality assurance, and the tracking of the impact of project performance (Auma 2014). Excellent quality performance can have the following benefits (Nassar 2009b):

1. Increase the organization's ability to provide its services
2. Increase customer satisfaction and thus increase the chances of repeat business
3. Reduce rework and improve construction efficiency

5.4 Team Satisfaction Performance Indicator

Human factors have a major impact on the quality of projects and their successful completion. Organizing and maintaining high-performance teams in today's competitive environment are two challenging tasks. Team members must support each other and communicate openly and clearly (Nassar 2009b). Job satisfaction indicates employees' desire and readiness for working in an organization. Accordingly, employees with a positive attitude are often productive. An attitude is the expression of feelings about people, objects, activities, and events. A wrong attitude can lead to poor performance and sabotage. Client satisfaction is a key indicator of project success. Thus, managers need to be aware of the team members' satisfaction levels so as to find a way to help them integrate and collaborate effectively. The TSPI is evaluated according to CE2004 based on 4 areas:

1. The extent to which team members influence their jobs
2. Wages and working conditions
3. Respect from department managers/supervisors
4. Sense of success at work

5.5 Client Satisfaction Performance Indicator

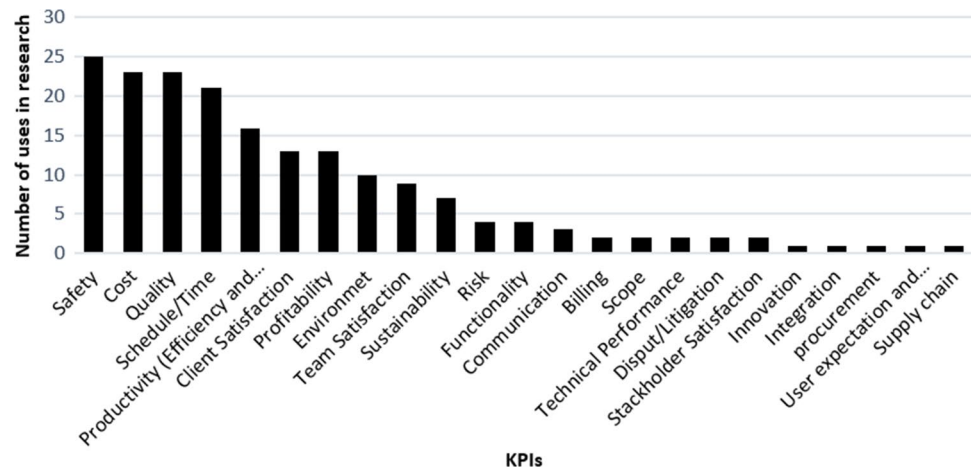
Success of a project depends on the company's ability to meet client needs so as to satisfy them. Client satisfaction can be achieved by providing quality products and services

Table 1 Key performance indicators examined in previous research

Row	References	Key performance indicators examined
1	Miller (1984)	Profitability, productivity
2	Bernolak (1997)	Quality
3	Ren (2000)	Productivity
4	Chan et al. (2002)	Time/schedule, cost, quality, profitability, safety, environment, team satisfaction, client satisfaction, productivity, technical performance, operation
5	Xiao and Proverbs (2003)	Time/schedule, cost, quality, profitability, safety, environment, client satisfaction, sustainability
6	Grünberg (2004)	Profitability, productivity
7	Soetanto and Proverbs (2004)	Client satisfaction
8	Chan and Chan (2004)	Time/schedule, cost, quality, profitability, safety, environment, team satisfaction, client satisfaction, productivity, technical performance, satisfaction, user expectations
9	Tangen (2005)	Profitability, productivity
10	Park (2006)	Productivity
11	Dawood et al. (2006)	Time, cost, safety, client satisfaction, communication, productivity
12	Bottazzi et al. (2008)	Profitability, productivity
13	Nasirzadeh et al. (2008)	Time, cost, quality, risk
14	Cousins et al. (2008)	Supply chain
15	Rankin et al. (2008)	Time, cost, quality, safety, environment, range, innovation, sustainability
16	Skibniewski and Ghosh (2009)	Time, cost, profitability, safety, client satisfaction, productivity
17	Bou-Llusar et al. (2009)	Quality
18	Nassar (2009a)	Time, cost, quality, profitability, safety, team satisfaction, client satisfaction, billing
19	Al-Humaidi and Tan (2010)	Safety
20	Wang et al. (2010)	Time, cost, quality, team satisfaction, scope, communication, integration, risk, procurement
21	Ali and Rahmat (2010)	Time, cost, quality, safety, customer satisfaction, operational
22	Ogunlana (2010)	Time, cost, quality, safety, stakeholder satisfaction, productivity, disputes/litigation
23	Pekuri et al. (2011)	Productivity
24	Cha and Kim (2011)	Time, cost, quality, safety, environment, productivity, risk
25	Rezaei et al. (2011)	Quality
26	Xu et al. (2012)	Safety
27	Ghoddousi and Hosseini (2012)	Productivity
28	Wanberg et al. (2013)	Quality, safety
29	Araújo et al. (2013)	Sustainability
30	Fung and Siow (2013)	Safety
31	Fang and Wu (2013)	Team satisfaction
32	Chovichien and Nguyen (2013)	Time, cost, quality, safety, environment, team satisfaction, client satisfaction, communication, productivity, technical, operational performance, user expectations, and satisfaction
33	Auma (2014)	Time, cost, quality
34	Nassar and AbouRizk (2014)	Time, cost, quality, profitability, safety, team satisfaction, customer satisfaction, billing
35	Fang et al. (2015)	Safety
36	Silvius and Schipper (2016)	Sustainability
37	Silva et al. (2016)	Time, cost, quality, profitability, safety, environment, team satisfaction, client satisfaction
38	Omar and Fayek (2016)	Time, cost, quality, safety, customer satisfaction, productivity
39	Wibowo et al. (2017)	Cost
40	Demirkesen and Ozorhon (2017a)	Time, cost, quality, safety
41	Leon et al. (2017)	Time, cost, quality, profitability, safety, environment, team satisfaction, client satisfaction
42	Demirkesen and Ozorhon (2017b)	Integration, scope, time, cost, quality, human resources, communications, risk, procurement, stakeholder, safety, environmental, financial; and claim
43	Xue et al. (2018)	Cost
44	Cha and Kim (2018)	Time, cost, quality, safety, environment, sustainability, productivity
45	Mohammadi et al. (2018)	Safety
46	Tripathi et al. (2019)	Profitability and asset management; satisfaction of key stakeholders; predictability of cost and time; environment, health and safety

Table 1 (continued)

Row	References	Key performance indicators examined
47	Keenan and Rostami (2019)	Quality
48	Winge et al. (2019)	Safety
49	Pham and Kim (2019)	Sustainability
50	Kang et al. (2020)	Safety
51	Sanni-Anibire et al. (2020)	Safety; risk

Fig. 6 Ranking of performance indicators

to it. Besides, client satisfaction has turned into a challenging issue in the construction industry over the past few decades. Clients in the construction sector often feel dissatisfied with many aspects, including additional project costs, completion delays, and poor quality of products delivered (Ali and Rahmat 2010). Findings show that construction companies can increase their profits by 100% as long as they can retain 5% or more of their existing clients. Therefore, client satisfaction is one of the key performance indicators for the success of construction projects (Soetanto and Proverbs 2004).

1. The following are six factors that affect customer satisfaction:
2. Project program control,
3. Project equivalent to customer budget
4. Quality of work
5. Customer-oriented response
6. Effective communication
7. Respond to complaints

5.6 Productivity Performance Indicator

Productivity is defined as the ratio of the amount of the output to the amount of the input in producing a deliverable. The output includes products and services, and the input includes materials, labor, assets, and energy.

Productivity means the amount and quality of the products produced from the available resources (Borcherding et al. 1986). The European National Association of Productivity Centers defines productivity as "the efficiency and effectiveness of products and services delivered." "Efficiency" can be explained as a measure of "doing things right" or using suitable resources for achieving the desired results (Bernolak 1997). On the other hand, "effectiveness" is defined as "doing the right thing," which indicates the extent to which client needs are met. In fact, effectiveness is related to the desired goal, yet efficiency is associated with the process (Neely et al. 1995).

Productivity assessment includes information gathered on specific tasks, performance in the field, and related working hours in a given period. Overtime, change orders, material management, as well as climate and human factors have been identified as the main factors affecting productivity. Intense competition in the construction industry causes all stakeholders to work so as to improve productivity. To improve productivity, it is necessary to study positive and negative factors affecting it (Ghoddousi and Hosseini 2012).

5.7 Profitability Performance Indicator

Profitability is one of the most important factors for business survival and an important goal to be achieved in production. This concept refers to the ability of a company or a project to

make a profit in a specific period of time (Zhang et al. 2015). It is an effective economic factor in demonstrating project performance. Profitability is often confused with productivity. The difference between these two concepts is that the former considers monetary effects, while the latter refers to a real process that takes place among purely physical phenomena. Profitability, like productivity, is a relationship between the input and the output; however, in the case of profitability, the relationship is monetary, so the impact of cost factors is included (Tangen 2005).

Miller described the relationship between the two in 1984 (Miller 1984):

$$\text{Profitability} = \text{Productivity} + \text{Price Recovery}$$

5.8 Safety Performance Indicator

The construction industry is among the riskiest industries, due to its unique, dynamic, and temporary nature (Fang and Wu 2013). In fact, disastrous effects of construction accidents in this industry are much higher than those in other industries (Mohammadi et al. 2018). Research shows that construction workers are three times as likely to die and twice as likely to be injured as workers in other industries (Sousa and Teixeira 2004). This makes us arrive at the conclusion that both managerial and economic aspects, including HR issues, time management, project management, project costs, humanitarian aspects, as well as ethical and social concerns about workers' welfare and safety, highlight the importance of improving performance. In addition, safety standards must be considered an important part of measures adopted in the construction industry (Mohammadi et al. 2018). Safety performance indicators (SPIs) indicate how work activities are performed without interruption.

5.9 Environmental Performance Indicator

Construction projects can have a major negative impact on the environment. Nowadays, the construction industry is more responsible for protecting the environment and minimizing damage. Not only does the construction industry consume resources, such as raw materials and energy, but it also generates wastes and hazardous air pollutants (Polster et al. 1996). According to past research, the environmental assessment of construction sites is one of the essential tools for the parties involved in a construction organization (Ren 2000). The environmental performance index (EPI) is considered as an evaluation criterion for business success in the long run and for the health of all organisms. The use of the EPI is economically and environmentally effective in measuring and improving productivity.

- Economic benefits: cost savings from methods such as energy efficiency and waste reduction
- Environmental benefits: reduction of fuel, CO₂ and other emissions that may cause global warming or air pollution

Some suggestions have been made, over time, about studying and evaluating environmental impacts, including energy consumption, dust and gas emissions, noise generation, waste generation, water discharge, use of water resources, unnecessary building uses, pollution caused by building materials, land use, and natural resources (Tam et al. 2006).

The use of key performance indicators in any research and projects for evaluating, predicting, or improving performance requires the determining of the relationships between indicators to calculate and quantify them. Table 1 demonstrates quantification relationships between the most important indicators identified in this study. Besides, all 100 studies reviewed are given in Table 1 to determine which index is used in each study. Upon referring to this table, the researcher will be able to identify and review all the sources related to each index (Table 2).

6 Interrelationships Key Performance Indicators

Figure 7 shows the major performance indicators, interactions between them, relationships between these factors, and reasons for their creation. As already mentioned in the previous sections, performance indicators have complex and interactive relationships. Most of these relationships and influences were determined based on the studies mentioned in the research literature and relevant reputable authorities through interviews with experts.

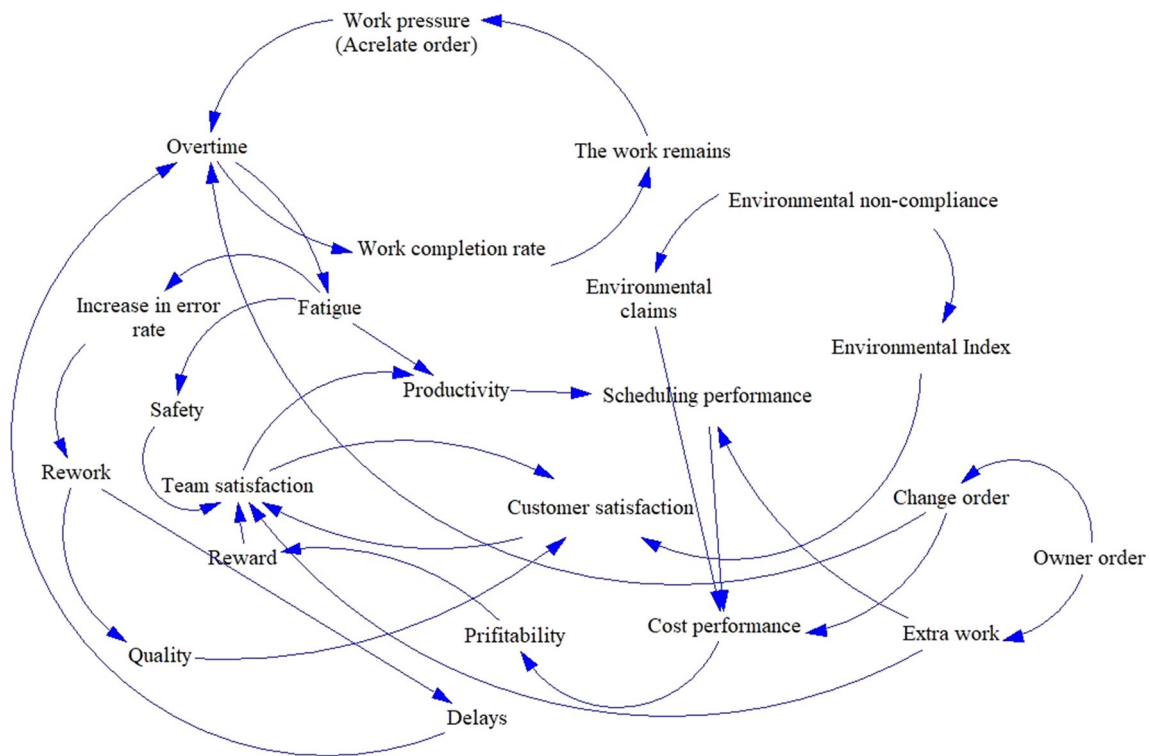
For example, employee fatigue caused by overtime affects safety and leads to rework. On the one hand, rework affects client satisfaction through the quality index; on the other, safety affects client satisfaction by affecting team satisfaction. In addition, an increase in productivity reduces costs and exerts a positive effect on cost performance. Productivity affects project scheduling performance as well. Poor program performance increases costs, thereby exerting a negative effect on cost performance. Costs affect project profitability, thereby affecting the profit margin. Besides, an increase in profits increases employee rewards, which in turn raises project team satisfaction. Delays and poor program performance may affect client satisfaction. In addition, client satisfaction and employee satisfaction have a bilateral relationship with and a synergic effect on each other. Rework is an important factor affecting costs and scheduling in construction projects. In addition, it can lead to the lowering of employee morale, thereby causing fatigue, reduced

Table 2 Quantitative relationships and sources of KPI

KPIs	Methodology to compute index	References
Cost performance index	$CPI = BCWP/ACWP$	Chan et al. (2002), Xiao and Proverbs (2003), Chan and Chan (2004), Dawood et al. (2006), Nasirzadeh et al. (2008), Rankin et al. (2008), Skibniewski and Ghosh (2009), Nassar (2009a), Wang et al. (2010), Ali and Rahmat (2010), Ogunlana (2010), Cha and Kim (2011), Chovichien and Nguyen (2013), Auma (2014), Nassar and AbouRizk (2014), Silva et al. (2016), Omar and Fayek (2016), Wibowo et al. (2017), Demirkesen and Ozorhon (2017a), Leon et al. (2017), Demirkesen and Ozorhon (2017b), Xue et al. (2018), Cha and Kim (2018), Mohammadi et al. (2018), Tripathi et al. (2019)
Schedule performance index	$SPI(t) = ES/AT$	Chan et al. (2002), Xiao and Proverbs (2003), Chan and Chan (2004), Dawood et al. (2006), Nasirzadeh et al. (2008), Rankin et al. (2008), Skibniewski and Ghosh (2009), Nassar (2009a), Wang et al. (2010), Ali and Rahmat (2010), Ogunlana (2010), Cha and Kim (2011), Chovichien and Nguyen (2013), Auma (2014), Nassar and AbouRizk (2014), Silva et al. (2016), Omar and Fayek (2016), Demirkesen and Ozorhon (2017a), Leon et al. (2017), Demirkesen and Ozorhon (2017b), Cha and Kim (2018), Tripathi et al. (2019)
Quality performance index	$CFRI = \frac{\text{total direct plus indirect cost of rework performed in the field}}{\text{total field construction phase cost}}$	Bernolak (1997), Chan et al. (2002), Xiao and Proverbs (2003), Chan and Chan (2004), Nasirzadeh et al. (2008), Rankin et al. (2008), Bou-Llusar et al. (2009), Nassar (2009a), Wang et al. (2010), Ali and Rahmat (2010), Ogunlana (2010), Cha and Kim (2011), Rezaei et al. (2011), Wanberg et al. (2013), Chovichien and Nguyen (2013), Auma (2014), Nassar and AbouRizk (2014), Silva et al. (2016), Demirkesen and Ozorhon (2017a), Demirkesen and Ozorhon (2017b), Cha and Kim (2018), Keenan and Ros-tami (2019)
Profitability performance index	$PPI = ER/ACWP$	Miller (1984), Chan et al. (2002), Xiao and Proverbs (2003), Grünberg (2004), Chan and Chan (2004), Tangen (2005), Bottazzi et al. (2008), Skibniewski and Ghosh (2009), Nassar (2009a), Nassar and AbouRizk (2014), Silva et al. (2016), Leon et al. (2017), Tripathi et al. (2019)
Safety performance index	$SFPI = \frac{\text{number of recordable incidents} \times 200,000}{\text{total site work hours}}$	Chan et al. (2002), Xiao and Proverbs (2003), Chan and Chan (2004), Dawood et al. (2006), Rankin et al. (2008), Skibniewski and Ghosh (2009), Al-Humaidi and Tan (2010), Ali and Rahmat (2010), Ogunlana (2010), Cha and Kim (2011), Xu et al. (2012), Wanberg et al. (2013), Fung and Siow (2013), Chovichien and Nguyen (2013), Nassar and AbouRizk (2014), Fang et al. (2015), Silva et al. (2016), Omar and Fayek (2016), Demirkesen and Ozorhon (2017a), Leon et al. (2017), Demirkesen and Ozorhon (2017b), Cha and Kim (2018), Mohammadi et al. (2018), Tripathi et al. (2019), Winge et al. (2019), Kang et al. (2020), Sanni-Anibire et al. (2020)

Table 2 (continued)

KPIs	Methodology to compute index	References
Environment performance index	$EPI = \sum_{i=1}^6 (W_i \times N_i) \sum_{i=1}^6 W_i = 1.0$	Chan et al. (2002), Xiao and Proverbs (2003), Chan and Chan (2004), Rankin et al. (2008), Cha and Kim (2011), Chovichien and Nguyen (2013), Silva et al. (2016), Leon et al. (2017), Demirkesen and Ozorhon (2017b), Cha and Kim (2018), Tripathi et al. (2019)
Team satisfaction performance index	Based on four area concern (a scale from 1 to 10)	Chan et al. (2002), Chan and Chan (2004), Nassar (2009a), Wang et al. (2010), Fang and Wu (2013), Chovichien and Nguyen (2013), Nassar and AbouRizk (2014), Silva et al. (2016), Leon et al. (2017)
Client satisfaction performance index	$CSPI = \sum_{j=1}^7 (W_j \times N_j) \sum_{j=1}^7 W_j = 1.0$	Chan et al. (2002), Xiao and Proverbs (2003), Soetanto and Proverbs (2004), Chan and Chan (2004), Dawood et al. (2006), Skibniewski and Ghosh (2009), Nassar (2009a), Chovichien and Nguyen (2013), Silva et al. (2016), Leon et al. (2017)
Productivity performance index	$PPI = \frac{\text{total installed cost}}{\text{total actual man_hours worked}}$	Miller (1984), Ren (2000), Chan et al. (2002), Grünberg (2004), Chan and Chan (2004), Tangen (2005), Park (2006), Dawood et al. (2006), Bottazzi et al. (2008), Skibniewski and Ghosh (2009), Ogunlana (2010), Pekuri et al. (2011), Cha and Kim (2011), Ghoddousi and Hosseini (2012), Chovichien and Nguyen (2013), Omar and Fayek (2016), Cha and Kim (2018)

**Fig. 7** Causal diagram

productivity, and ultimately dissatisfaction. Redesign can affect the ultimate quality of the project, thereby leading to client satisfaction. An increase in the number of recordable accidents (unforeseen accidents) can affect team performance, thereby causing dissatisfaction among the team members and leading to client dissatisfaction. Failure to meet environmental commitments may lead to claims that affect project cost and schedule. Figure 7 shows these relationships. In addition, assumptions related to Fig. 7, which were obtained from previous studies and interviews with experts, are presented in Table 3.

7 Future Research Directions

In most cases, a simulation method per se cannot take into account different factors with different classifications, so the researcher has to combine two or more problem-solving and modeling methods. The greater the complexities of

a problem are, the more the need for a hybrid simulation method will be. Construction projects require robust and synthetic simulations because of their increasing complexity and progress. Therefore, for future research, the system dynamics method could be combined with other methods such as expert systems, case-based reasoning, and factor-based simulations to obtain more accurate and reality-based results. To obtain accurate information on the future performance of all types of projects, it is possible to provide a comprehensive assessment model of their performance, by taking into account all functional aspects and types of projects.

8 Conclusions

Project managers in the construction industry must be able to make reliable predictions of the future status of projects. However, given the complexity and progress of

Table 3 Research hypotheses

Influential variables	Influenced variables	Type of effect
The work remains	Work pressure (acceleration command)	+
Work pressure (acceleration command)	Overtime	+
Overtime	Work completion rate	+
Overtime	Fatigue	+
Work completion rate	The work remains	–
Fatigue	Efficiency	–
Fatigue	Safety	–
Increase in error rate	Rework	+
Rework	Quality	–
Rework	Delays	+
Safety	Team satisfaction	+
Team satisfaction	Efficiency	+
Efficiency	Scheduling performance	–
Customer satisfaction	Team satisfaction	+
Scheduling performance	Cost performance	+
Cost performance	Profitability	+
Profitability	Reward	+
Reward	Team satisfaction	+
Quality	Customer satisfaction	+
Delays	Overtime	+
Owner orders	Change command	+
Owner orders	Extra work	+
Change command	Cost performance	–
Extra work	Team satisfaction	–
Extra work	Scheduling performance	–
Environmental non-compliance	Environmental claims	+
Environmental non-compliance	Environmental index	–
Environmental claims	Cost performance	–
Environmental Index	Customer satisfaction	+

construction projects, such predictions require critical project information, used to evaluate project performance, thereby predicting it. In this study, the major measurement systems and frameworks were identified by reviewing the research literature on construction project performance. Balance scorecard is the most useful framework among performance measurement frameworks, in terms of performance evaluation due to its frequent use in recent years as well as its applicability to project-oriented organizations.

In this study, the key indicators used to evaluate the performance of construction projects were collected and reviewed through a comprehensive review of the relevant research literature. Next, they were prioritized after preparing the use diagram of these indicators in recent years. The most important indicators in order of priority used in research are safety, cost, quality, scheduling, productivity, client satisfaction, profitability, team satisfaction, environment, and sustainability. Among the authors who studied project performance management techniques in the literature, 52% of them considered the safety index as a key performance indicator in construction projects interacting with other indicators or alone, with 73% of whom having been related to the 10 past years. Subsequent indicators are cost and quality, which account for 40% of the research, with 63% of which having been done in the past 10 years. The timing/scheduling index has also been recognized as the 4th key performance indicator in recent years, with 45% of the studies have used it in interaction with other indicators or alone, and with 60% of which having been done in the past 10 years. The ranking for other performance indicators includes productivity with 36%, client satisfaction with 31%, profitability with 27%, environmental and team satisfaction with 20% each, and sustainability with 13%. The system dynamics approach was also presented as a dynamic approach to project management. Project management has been one of the successful users of the system dynamics approach in recent years. System dynamics is an approach to complex systems analysis and an effective tool for evaluating a system's ability to adjust and test new decisions that should be used for a more effective decision process. An application of this approach is to use the performance management of construction projects. The prediction and management of construction projects are complex and dynamic processes that should be used to study the system dynamics approach to accurately simulate the complex interactions among key performance indicators and provide a complete work model. The importance of prediction and management of construction projects has been demonstrated by the successful application of the system dynamics approach to performance management in recent years as well as the analysis of models presented in this research.

References

- Adèr HJ (2008) Advising on research methods: a consultant's companion. Johannes van Kessel Publishing, Huizen
- Al-Humaidi H, Tan FH (2010) Construction safety in Kuwait. *J Perform Constr Facil* 24:70–77. [https://doi.org/10.1061/\(ASCE\)CF.1943-5509.0000055](https://doi.org/10.1061/(ASCE)CF.1943-5509.0000055)
- Ali AS, Rahmat I (2010) The performance measurement of construction projects managed by ISO-certified contractors in Malaysia. *J Retail Leis Prop* 9:25–35
- Al-Tabtabai H (1998) A framework for developing an expert analysis and forecasting system for construction projects. *Expert Syst Appl* 14:259–273
- Araújo C, Bragança L, Almeida MGD (2013) Sustainable construction key indicators. In: Portugal SB13-contribution of sustainable building to meet EU 20-20-20 Targets, 2013. pp 505–512
- Atkinson R (1999) Project management: cost, time and quality, two best guesses and a phenomenon, its time to accept other success criteria. *Int J Project Manag* 17:337–342
- Auma E (2014) Factors affecting the performance of construction projects in Kenya: a survey of low-rise buildings in Nairobi Central Business District The International. *J Bus Manag* 2:115
- Baccarini D (1999) The logical framework method for defining project success. *Project Manag J* 30:25–32
- Barraza GA, Back WE, Mata F (2004) Probabilistic forecasting of project performance using stochastic S curves. *J Constr Eng Manag* 130:25–32
- Beatham S, Anumba C, Thorpe T, Hedges I (2004) KPIs: a critical appraisal of their use in construction. *Benchmark: Int J* 11(1):93–117
- Bernolak I (1997) Effective measurement and successful elements of company productivity: the basis of competitiveness and world prosperity. *Int J Prod Econ* 52:203–213. [https://doi.org/10.1016/S0925-5273\(97\)00026-1](https://doi.org/10.1016/S0925-5273(97)00026-1)
- Borcherding JD, Palmeter S, Jansma G (1986) Work force management programs for increased productivity and quality work. In: EEI construction committee spring meetings
- Bottazzi G, Secchi A, Tamagni F (2008) Productivity, profitability and financial performance. *Ind Corp Change* 17:711–751. <https://doi.org/10.1093/icc/dtn027>
- Bou-Lluisar JC, Escrig-Tena AB, Roca-Puig V, Beltrán-Martín I (2009) An empirical assessment of the EFQM excellence model: evaluation as a TQM framework relative to the MBNQA Model. *J Oper Manag* 27:1–22. <https://doi.org/10.1016/j.jom.2008.04.001>
- Cha HS, Kim CK (2011) Quantitative approach for project performance measurement on building construction in South Korea. *KSCE J Civ Eng* 15:1319–1328. <https://doi.org/10.1007/s12205-011-1323-5>
- Cha HS, Kim KH (2018) Measuring project performance in consideration of optimal best management practices for building construction in South Korea. *KSCE J Civ Eng* 22:1614–1625. <https://doi.org/10.1007/s12205-017-0156-2>
- Chan AP, Chan AP (2004) Key performance indicators for measuring construction success. *Benchmark: Int J* 11:203–221. <https://doi.org/10.1108/14635770410532624>
- Chan AP, Scott D, Lam EW (2002) Framework of success criteria for design/build projects. *J Manag Eng* 18:120–128. [https://doi.org/10.1061/\(ASCE\)0742-597X\(2002\)18:3\(120\)](https://doi.org/10.1061/(ASCE)0742-597X(2002)18:3(120))
- Choi K, Bae D-H (2009) Dynamic project performance estimation by combining static estimation models with system dynamics. *Inf Softw Technol* 51:162–172. [https://doi.org/10.1061/\(ASCE\)0742-597X\(2002\)18:3\(120\)](https://doi.org/10.1061/(ASCE)0742-597X(2002)18:3(120))
- Chovichien V, Nguyen TA List of indicators and criteria for evaluating construction project success and their weight assignment. In: Proceedings of the 4th international conference on engineering,

- project, and production management (EPPM 2013), 2013. pp 130–150. <https://doi.org/https://doi.org/10.32738/CEPPM.201310.0011>.
- Cooper H (1986) The integrative research review: A systematic approach sage publications: Beverly Hills, 1984, 143 pp Educational Researcher 15:17–18
- Corovic R (2006) Why EVM is not good for schedule performance analyses (and how it could be...). The Measurable News, Winter
- Cousins PD, Lawson B, Squire B (2008) Performance measurement in strategic buyer-supplier relationships: the mediating role of socialization mechanisms. *Int J Oper Prod Manag* 28:238–258. <https://doi.org/10.1108/01443570810856170>
- Dawood N, Sikka S, Marasini R, Dean J Development of key performance indicators to establish the benefits of 4D planning'. In: Proceedings 22nd Annual ARCOM Conference, 2006. vol 6.
- De Wit A (1988) Measurement of project success. *Int J Project Manag* 6:164–170. [https://doi.org/10.1016/0263-7863\(88\)90043-9](https://doi.org/10.1016/0263-7863(88)90043-9)
- Demirkesen S, Ozorhon B (2017a) Impact of integration management on construction project management performance. *Int J Project Manag* 35:1639–1654
- Demirkesen S, Ozorhon B (2017b) Measuring project management performance: case of construction industry. *Eng Manag J* 29:258–277
- Ding Z, Yi G, Tam VW, Huang T (2016) A system dynamics-based environmental performance simulation of construction waste reduction management in China. *Waste Manag* 51:130–141. <https://doi.org/10.1016/j.wasman.2016.03.001>
- Eden C, Williams T, Ackermann F, Howick S (2000) The role of feedback dynamics in disruption and delay on the nature of disruption and delay (D&D) in major projects. *J Oper Res Soc* 51:291. <https://doi.org/10.1057/palgrave.jors.2600919>
- Excellence C (2006) UK construction industry key performance indicators: handbook CE. London, 14
- Fang D, Wu H (2013) Development of a Safety Culture Interaction (SCI) model for construction projects. *Saf Sci* 57:138–149. <https://doi.org/10.1016/j.ssci.2013.02.003>
- Fang D, Wu C, Wu H (2015) Impact of the supervisor on worker safety behavior in construction projects. *J Manag Eng* 31:04015001. [https://doi.org/10.1061/\(ASCE\)ME.1943-5479.0000355](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000355)
- Farghal SH, Everett JG (1997) Learning curves: accuracy in predicting future performance. *J Constr Eng Manag* 123:41–45
- Freeman M, Beale P (1992) Measuring project success. In: Project Management Institute
- Fung HP, Siow H (2013) Relationship between team satisfaction and project performance as perceived by project managers in Malaysia—A mixed methods study. *Open J Soc Sci Res OJSSR* 1:238–249. <https://doi.org/10.12966/OJSSR.12.02.2013>
- Ghoddousi P, Hosseini MR (2012) A survey of the factors affecting the productivity of construction projects in Iran. *Technol Econ Dev Econ* 18:99–116. <https://doi.org/10.3846/20294913.2012.661203>
- Grünberg T (2004) Performance improvement: Towards a method for finding and prioritising potential performance improvement areas in manufacturing operations. *Int J Product Perform Manag* 53:52–71. <https://doi.org/10.1108/17410400410509969>
- Henderson K (2007) Earned schedule a breakthrough, extension to earned value management. In: Proceedings of PMI global congress Asia Pacific
- Heravi G, Ilbeigi M (2012) Development of a comprehensive model for construction project success evaluation by contractors. *Eng Constr Arch Manag* 19:526–542
- Hwang B-G, Thomas SR, Haas CT, Caldas CH (2009) Measuring the impact of rework on construction cost performance. *J Constr Eng Manag* 135:187–198
- Ibbs W, Liu M (2005) System dynamic modeling of delay and disruption claims. *Cost Eng* 47:12
- Idrus A, Sodangi M, Husin MH (2011) Prioritizing project performance criteria within client perspective Research Journal of Applied Sciences. *Eng Technol* 3:1142–1151
- Ingle PV, Mahesh G (2020) Construction project performance areas for Indian construction projects. *Int J Constr Manag* 1–12
- Iram N, Khan B, Sahibzada UF, Ahmad MS (2016) Critical factors influencing the project success: An analysis of projects in manufacturing and construction industries in Punjab, Pakistan. *Int J Bus Stud Rev* 1:34–43
- Kang L, Wu C, Liao X, Wang B (2020) Safety performance and technology heterogeneity in China's provincial construction industry. *Saf Sci* 121:83–92. <https://doi.org/10.1016/j.ssci.2019.09.005>
- Keenan M, Rostami A (2019) The impact of quality management systems on construction performance in the North West of England. *Int J Constr Manag*. <https://doi.org/10.1080/15623599.2019.1590974>
- Kennerley M, Neely A (2002) A framework of the factors affecting the evolution of performance measurement systems. *Int J Oper Prod Manag* 22:1222–1245. <https://doi.org/10.1108/01443570210450293>
- Leon H, Osman H, Georgy M, Elsaid M (2017) System dynamics approach for forecasting performance of construction projects. *J Manag Eng* 34:04017049. [https://doi.org/10.1061/\(ASCE\)ME.1943-5479.0000575](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000575)
- Leong TK, Zakuan N, Mat Saman MZ, Ariff M, Md S, Tan CS (2014) Using project performance to measure effectiveness of quality management system maintenance and practices in construction industry. *Sci World J*. <https://doi.org/10.1155/2014/591361>
- Li J (2004) Web-based integrated project control. *Constr Manag Econ* 22(1):35–46
- Lim C, Mohamed MZ (1999) Criteria of project success: an exploratory re-examination. *Int J Project Manag* 17:243–248. [https://doi.org/10.1016/S0263-7863\(98\)00040-4](https://doi.org/10.1016/S0263-7863(98)00040-4)
- Lipke W (2003) Schedule is different The Measurable News 31:31–34
- Love PE, Mandal P, Li H (1999) Determining the causal structure of rework influences in construction. *Constr Manag Econ* 17:505–517. <https://doi.org/10.1080/014461999371420>
- Miller DM (1984) Profitability = productivity + price recovery. *Harvard Bus Rev* 62:145–153
- Mohammadi A, Tavakolan M (2019) Modeling the effects of production pressure on safety performance in construction projects using system dynamics. *J Safety Res* 71:273–284
- Mohammadi A, Tavakolan M, Khosravi Y (2018) Factors influencing safety performance on construction projects: a review. *Saf Sci* 109:382–397. <https://doi.org/10.1016/j.ssci.2018.06.017>
- Munns AK, Bjeirmi BF (1996) The role of project management in achieving project success. *Int J Project Manag* 14:81–87. [https://doi.org/10.1016/0263-7863\(95\)000577](https://doi.org/10.1016/0263-7863(95)000577)
- Nabi MA, El-adaway IH, Dagli C (2020) A system dynamics model for construction safety behavior. *Procedia Comput Sci* 168:249–256
- Naeni LM, Shadrokh S, Salehipour A (2014) A fuzzy approach for the earned value management. *Int J Project Manag* 32:709–716
- Nasirzadeh F, Afshar A, Khanzadi M (2008) System dynamics approach for construction risk analysis. *Int J Civ Eng* 6:120–131
- Nassar NK (2009a) An integrated framework for evaluation of performance of construction projects. Paper presented at PMI® Global Congress 2009—North America, Orlando, FL. Project Management Institute, Newtown Square, PA
- Nassar NK (2009b) An integrated framework for evaluation of performance of construction projects. In: PMI Global congress, north america: project management institute
- Nassar N, AbouRizk S (2014) Practical application for integrated performance measurement of construction projects. *J Manag Eng* 30:04014027. [https://doi.org/10.1061/\(ASCE\)ME.1943-5479.0000287](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000287)

- Neely A, Gregory M, Platts K (1995) Performance measurement system design: a literature review and research agenda. *Int J Oper Prod Manag* 15:80–116
- Neely A, Mills J, Platts K, Richards H, Gregory M, Bourne M, Kennerley M (2000) Performance measurement system design: developing and testing a process-based approach. *Int J Oper Prod Manag* 20:1119–1145. <https://doi.org/10.1108/01443570010343708>
- Oates BJ, Capper G Using systematic reviews and evidence-based software engineering with masters students. In: 13th International Conference on Evaluation and Assessment in Software Engineering (EASE) 13, 2009. pp 1–9
- Ogunlana SO (2010) Beyond the ‘iron triangle’: Stakeholder perception of key performance indicators (KPIs) for large-scale public sector development projects. *Int J Project Manag* 28:228–236. <https://doi.org/10.1016/j.ijproman.2009.05.005>
- Ogunlana SO, Li H, Sukhera FA (2003) System dynamics approach to exploring performance enhancement in a construction organization. *J Constr Eng Manag* 129:528–536. [https://doi.org/10.1061/\(ASCE\)0733-9364\(2003\)129:5\(528\)](https://doi.org/10.1061/(ASCE)0733-9364(2003)129:5(528))
- Omar MN, Fayek AR (2016) Modeling and evaluating construction project competencies and their relationship to project performance. *Autom Constr* 69:115–130. <https://doi.org/10.1016/j.autcon.2016.05.021>
- Owolabi James D, Amusan Lekan M, Oloke C, Olusanya O, Tunji-Olayeni P (2014) Causes and effect of delay on project construction delivery time. *Int J Educ Res* 2:197–208
- Parfitt MK, Sanvido V (1993) Checklist of critical success factors for building projects. *J Manag Eng* 9:243–249. [https://doi.org/10.1061/\(ASCE\)9742-597X\(1993\)9:3\(243\)](https://doi.org/10.1061/(ASCE)9742-597X(1993)9:3(243))
- Parikh DV, Phugat S (2019) Performance management in road construction project. *Int Res J Eng Technol (IRJET)* 6(4)
- Park H-S (2006) Conceptual framework of construction productivity estimation. *KSCE J Civ Eng* 10:311–317. <https://doi.org/10.1007/BF02830084>
- Pekuri A, Haapasalo H, Herrala M (2011) Productivity and performance management—managerial practices in the construction industry. *Int J Perform Meas* 1:39–58
- Pham H, Kim S-Y (2019) The effects of sustainable practices and managers’ leadership competences on sustainability performance of construction firms. *Sustain Prod Consum* 20:1–14. <https://doi.org/10.1016/j.spc.2019.05.003>
- Polster B, Peuportier B, Sommereux IB, Pedregal PD, Gobin C, Durand E (1996) Evaluation of the environmental quality of buildings towards a more environmentally conscious design. *Sol Energy* 57:219–230
- Rankin J, Fayek AR, Meade G, Haas C, Manseau A (2008) Initial metrics and pilot program results for measuring the performance of the Canadian construction industry. *Can J Civ Eng* 35:894–907. <https://doi.org/10.1139/L08-018>
- Ren X (2000) Development of environmental performance indicators for textile process and product. *J Clean Prod* 8:473–481. [https://doi.org/10.1016/S0959-6526\(00\)00017-2](https://doi.org/10.1016/S0959-6526(00)00017-2)
- Rezaei A, Çelik T, Baalousha Y (2011) Performance measurement in a quality management system. *Scientia Iranica* 18:742–752. <https://doi.org/10.1016/j.scient.2011.05.021>
- Sanni-Anibire MO, Mahmoud AS, Hassanain MA, Salami BA (2020) A risk assessment approach for enhancing construction safety performance. *Saf Sci* 121:15–29. <https://doi.org/10.1016/j.ssci.2019.08.044>
- Sanvido V, Grobler F, Parfitt K, Guvenis M, Coyle M (1992) Critical success factors for construction projects. *J Constr Eng Manag* 118:94–111
- Shen L, Bao Q, Ip S (2000) Implementing innovative functions in construction project management towards the mission of sustainable development. In: Proceedings of the millennium conference on construction project management, Hong Kong, October, pp 77–84, 2000
- Shenhar AJ, Wideman RM (1996) Improving PM: Linking success criteria to project type. In: Project management institute, symposium, creating canadian advantage through project management, Calgary, Canada, 1996.
- Silva G, Warnakulasooriya B, Arachchige B (2016) Criteria for construction project success: a literature review. <https://doi.org/https://doi.org/10.2139/ssrn.2910305>
- Silvius AG, Schipper R (2016) Exploring the relationship between sustainability and project success—conceptual model and expected relationships. *Int J Inf Syst Project Manag* 4:5–22. <https://doi.org/10.12821/IJISPM40301>
- Skibniewski MJ, Ghosh S (2009) Determination of key performance indicators with enterprise resource planning systems in engineering construction firms. *J Constr Eng Manag* 135:965–978. [https://doi.org/10.1061/\(ASCE\)0733-9364\(2009\)135:10\(965\)](https://doi.org/10.1061/(ASCE)0733-9364(2009)135:10(965))
- Soetanto R, Proverbs DG (2004) Intelligent models for predicting levels of client satisfaction. *J Constr Res* 5:233–253. <https://doi.org/10.1142/S1609945104000164>
- Sousa S, Teixeira J (2004) Prevention measures to reduce risk of falling from heights. In: IX National Symposium of ISMAI, 2004. p 15
- Sterman JD (1992) System dynamics modeling for project management Unpublished manuscript, Cambridge, MA 246
- Sterman J (2000) Business dynamics: systems thinking and modeling for a complex world: Jeffrey J Shelstad. Indianapolis
- Tam VW, Tam C, Zeng S, Chan K (2006) Environmental performance measurement indicators in construction. *Build Environ* 41:164–173
- Tangen S (2005) Demystifying productivity and performance. *Int J Prod Perform Manag* 54:34–46. <https://doi.org/10.1108/17410400510571437>
- Tripathi KK, Hasan A, Neeraj Jha K (2019) Evaluating performance of construction organizations using fuzzy preference relation technique. *Int J Constr Manag*. <https://doi.org/10.1080/15623599.2019.1613210>
- Vyas G, Kulkarni S (2013) Performance indicators for construction project. *Int J Adv Electric Electron Eng (IAEEE)* 2(1)
- Walworth T, Yearworth M, Davis J, Davies P Early estimation of project performance: the application of a system dynamics rework model. In: 2013 IEEE international systems conference (SysCon), 2013. IEEE, pp 204–208. <https://doi.org/https://doi.org/10.1109/SysCon.2013.6549883>
- Wanberg J, Harper C, Hallowell MR, Rajendran S (2013) Relationship between construction safety and quality performance. *J Constr Eng Manag* 139:04013003. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0000732](https://doi.org/10.1061/(ASCE)CO.1943-7862.0000732)
- Wang Q, El-Gafy M, Zha J Bi-level framework for measuring performance to improve productivity of construction enterprises. In: Construction research congress 2010: innovation for reshaping construction practice, 2010. pp 970–979. [https://doi.org/https://doi.org/10.1061/41109\(373\)97](https://doi.org/https://doi.org/10.1061/41109(373)97)
- Ward S, Curtis B, Chapman C (1991) Objectives and performance in construction projects. *Constr Manag Econ* 9:343–353. <https://doi.org/10.1080/01446199100000027>
- Wells Jr W (1998) From the Editor (editorial on project success and failure) *Project Management Journal*
- Wibowo MA, Astana INY, Rusdi H (2017) Dynamic modelling of the relation between bidding strategy and construction project performance. *Procedia Engineering* 171:341–347. <https://doi.org/10.1016/j.proeng.2017.01.342>
- Winge S, Albrechtsen E, Arnesen J (2019) A comparative analysis of safety management and safety performance in twelve construction projects. *J Saf Res*. <https://doi.org/10.1016/j.jsr.2019.09.015>

- Xiao H, Proverbs D (2003) Factors influencing contractor performance: an international investigation Engineering. Constr Arch Manag 10:322–332. <https://doi.org/10.1108/09699980310502937>
- Xu Y, Wang T, Song B, Pang L, Xie Y (2012) Forecast of safety situation in construction industry based on composite model. Procedia Eng 45:119–124. <https://doi.org/10.1016/j.proeng.2012.08.131>
- Xue H, Zhang S, Su Y, Wu Z, Yang RJ (2018) Effect of stakeholder collaborative management on off-site construction cost performance. J Clean Prod 184:490–502. <https://doi.org/10.1016/j.jclepro.2018.02.258>
- Yun S, Choi J, de Oliveira DP, Mulva SP (2016) Development of performance metrics for phase-based capital project benchmarking. Int J Project Manage 34:389–402. <https://doi.org/10.1016/j.ijproman.2015.12.004>
- Zhang Q, Xu Z, Feng T, Jiao J (2015) A dynamic stochastic frontier model to evaluate regional financial efficiency: Evidence from Chinese county-level panel data. Eur J Oper Res 241:907–916